FinDS-AgentBench: A Pilot Benchmark for Financial Data-Science Agents

FinDS-AgentBench Contributors

Abstract

We introduce FinDS-AgentBench, a benchmark for evaluating AI agents that perform end-to-end financial data-science research under temporal, leakage, reproducibility, and decision-quality constraints. The current pilot release contains 9 tasks, 9 runnable tasks, and 3 tracks. The repeated-run protocol covers 8 tasks and 35 task-system cells with fixed seeds and release-owned artifacts. In the pilot overall-score paired comparison against the best baseline per task, agents are higher in 0 cases, baselines are higher in 6 cases, and 2 cases are ties. The release ships task cards, data manifests, reference results, uncertainty tables, manual-audit artifacts, and a deterministic release reproducibility check.

1 Introduction

Financial machine-learning research is unusually sensitive to timestamp discipline, point-in-time data availability, target leakage, validation design, transaction costs, and narrative overclaiming. Generic data-science and tool-use benchmarks do not isolate these failure modes. FinDS-AgentBench is designed to evaluate whether agents can produce auditable financial research artifacts rather than merely plausible notebooks.

Contributions.

1. A pilot benchmark release with task specifications, public data manifests, task cards, and evaluation cards.
2. A command-based agent and baseline protocol with repeated runs, fixed seeds, clean release builds, and manifest validation.
3. Publication-facing reference results, uncertainty estimates, paired agent-vs-baseline comparisons, and paper-ready tables.
4. A manual-audit workflow covering temporal protocol correctness, leakage awareness, evidence use, claim discipline, and reproducibility trace completeness.

2 Related Work

FinDS-AgentBench sits at the intersection of agent benchmarks, machine-learning experimentation benchmarks, finance-domain LLM evaluation, and leakage-aware temporal validation. General agent and software-engineering benchmarks such as [11, 7] emphasize multi-step execution, but do not target financial ML research validity. ML-oriented agent benchmarks such as [5, 1, 3, 2] evaluate experimentation, engineering, or open-ended research automation. FinDS-AgentBench adopts

Work	Neighbor Area	FinDS-AgentBench distinction
AgentBench [11]	General agent benchmark	Broad agent tasks; no financial ML protocol or point-in-time validity gates.
SWE-bench [7]	Software-engineering agent benchmark	Software repair rather than financial research notebooks and temporal holdouts.
DS-1000 [9]	Data-science agent benchmark	Code-generation tasks rather than full financial ML research artifacts.
InfiAgent-DABench [4]	Data-science agent benchmark	General data analysis without finance-specific leakage and holdout discipline.
MLAgentBench [5]	ML agent benchmark	ML experimentation without finance-specific data calendars or claim audit.
MLE-bench [1]	ML agent benchmark	Kaggle-style ML engineering rather than financial temporal information discipline.
ScienceAgentBench [3]	General agent benchmark	Scientific workflows without market data, finance metrics, or investment-claim audit.
MLR-Bench [2]	ML agent benchmark	Open-ended ML research without finance leakage, private holdouts, or model-risk framing.
FinanceBench [6]	Finance-domain agent/LLM benchmark	Financial QA rather than executable financial ML research artifacts.
FinBen [14]	Finance-domain agent/LLM benchmark	Broad financial LLM skills rather than validity-gated research workflows.
Profit Mirage / FinLake-Bench [10]	Finance-domain agent/LLM benchmark	Leakage-aware financial agents, but not reproducible research notebooks and audit cards.
TemporalBench [13]	Temporal reasoning benchmark	Time-series reasoning without financial ML artifacts and private temporal scoring.
FinToolBench [12]	Finance-domain agent/LLM benchmark	Tool invocation rather than leakage-safe financial ML notebooks.
FinMCP-Bench [16]	Finance-domain agent/LLM benchmark	MCP tool use rather than end-to-end financial ML research workflows.
WorkstreamBench [15]	Finance-domain agent/LLM benchmark	Spreadsheet workflows rather than temporal prediction and leakage-safe scoring.
BlueFin [8]	Finance-domain agent/LLM benchmark	Spreadsheet tasks rather than executable financial ML notebooks and holdouts.

Table 1: Related benchmark positioning. FinDS-AgentBench targets the missing intersection of end-to-end financial ML research artifacts, temporal information discipline, leakage resistance, reproducibility, and writeup validity.

their emphasis on executable artifacts while adding point-in-time finance data, temporal holdouts, leakage scanning, and writeup audit. Finance benchmarks such as [6, 14, 10, 12, 16, 15, 8] cover financial QA, broad financial LLM capabilities, leakage in financial agents, tool use, and spreadsheet workflows. These are adjacent but do not provide an end-to-end financial ML research benchmark with point-in-time task cards, private temporal labels, repeated-run uncertainty, reproducibility gates, and narrative-discipline scoring. Data-science and data-analysis benchmarks such as [9, 4] motivate reliable code execution and analysis-task evaluation. FinDS-AgentBench extends that execution focus to financial research protocols where model validation, data timestamp availability, and claims about economic usefulness must be evaluated jointly. Temporal-reasoning benchmarks such as [13] motivate time-aware evaluation. FinDS-AgentBench narrows that lens to finance, where point-in-time information availability and leakage-safe validation are central rather than incidental constraints.

3 Benchmark Design

The pilot release spans event-aware time-series reasoning, predictive financial ML, research replication and audit. Each task specification identifies the prediction timestamp, allowed public information, target horizon, temporal split, required submission artifacts, primary metric, and leakage constraints. The benchmark separates predictive performance from artifact validity so that a high private-holdout score does not erase reproducibility, leakage, or methodology failures.

4 Pilot Release

The release manifest records 9 tasks and 9 runnable tasks. The repeated-run pilot protocol covers 8 tasks, excluding the leakage-audit task from the predictive baseline/agent sweep because it uses an audit-style scoring surface rather than a standard prediction holdout.

5 Evaluation Protocol

The combined pilot protocol evaluates 27 baseline task-system cells and 8 agent task-system cells. Observed run counts per cell are 3. Baselines include naive and classical models; the bundled example agents are environment-wrapped systems that select among public candidate strategies

Task	Agent vs. Baseline	n	Agent	Baseline	Δ (p_{sign})
Front-End Spread	Front-End Sweep Env-Agent / LogReg	3	0.543	0.543	0.000 (–)
Synthetic Event	Event Rule Env-Agent / Event Rule	3	0.637	0.637	0.000 (–)
Synthetic Market	Market Sweep Env-Agent / Momentum	3	0.511	0.521	-0.010 (0.250)
USD AFE vs. EME	USD Sweep Env-Agent / Extra Trees	3	0.507	0.529	-0.022 (0.250)
USD Broad	USD Sweep Env-Agent / Prev-Day	3	0.502	0.510	-0.008 (0.250)
Treasury 10Y-2Y Curve	Treasury 10Y-2Y Sweep Env-Agent / LogReg	3	0.501	0.530	-0.029 (0.250)
Treasury 10Y-3M Curve	Treasury 10Y-3M Sweep Env-Agent / Extra Trees	3	0.517	0.533	-0.016 (0.250)
Treasury 10Y Direction	Treasury 10Y Sweep Env-Agent / Extra Trees	3	0.504	0.506	-0.002 (1.000)

Table 2: Paired agent-minus-best-baseline comparison for Overall. Baselines are selected separately within each task by completed-run mean. The sign-test p-value is exact and two-sided.

before writing holdout predictions and research artifacts. The external-agent readiness gate is `not_ready_no_external_agents` with 0 completed external agent configurations.

Statistical reporting. We report repeated-run uncertainty from the combined protocol run table `reports/release_runs/pilot_protocol_run_results.csv`. The reported metrics are `score.overall_score`, `score.balanced_accuracy`, `score.roc_auc`. For each task, system, run type, and metric, completed runs with numeric metric values are summarized by the sample mean, sample standard deviation, standard error, and a two-sided 95% t-interval. Single-run groups are reported with zero-width intervals rather than extrapolated uncertainty. For agent-vs-baseline comparisons, the best baseline is selected independently for each task and metric using the completed-run mean. Agent and baseline runs are paired by `trace.repeat_index`, falling back to `trace.seed` only when no repeat index is available. We report the paired agent-minus-baseline mean difference, its t-interval, and an exact two-sided sign-test p-value after dropping zero differences. Because the pilot uses a small repeated-run count, these statistics are treated as transparent uncertainty and regression artifacts rather than definitive significance claims.

6 Results

Table 2 compares each agent with the best completed-run baseline for the same task. Appendix Table 3 reports repeated-run overall-score uncertainty for all task-system cells.

7 Manual Audit and Validity Checks

The pilot manual-audit bundle currently includes 6 cases across 3 reviewed tasks. Its official status is `seed_author_adjudication_only`; independent-overlap agreement status is `pairwise_agreement_available`, with exploratory status `pairwise_agreement_available`. The reviewer-readiness gate is `ready_for_submission_claims` with 1 completed independent reviewer packets. This makes the current audit layer suitable as a seed adjudication workflow, while a submission-ready paper should add an independent second-reviewer packet.

8 Qualitative Failure Examples

The manual audit illustrates why predictive scores and automatic artifact validators are insufficient for financial research-agent evaluation. The examples below are generated from the seed adjudicated audit subset and preserve task, run, and artifact references.

1. **Case 1:** `pilot_event_rule_baseline_release_001`. **Task/system:** `synthetic_event_response_v0` / `event_rule_baseline` (baseline, run label `release_pilot_event_001_seed_23`). **Audit label:** `minimally_defensible`, total score 6. **Low-scoring dimensions:** `temporal_protocol_correctness`, `quantitative_evidence_use`, `baseline_comparison_or_counterfactual_context`, `reproducibility_trace_completeness`. **Primary findings:** Strong predictive score is not accompanied by quantitative narrative support; Temporal protocol description is brief; No comparator or ablation context appears in the writeup. **Evidence:** No ablation, heuristic comparison, or simpler event rule is discussed. **Artifact reference:** `runs/suites/pilot_baselines_v0/synthetic_event_response_v0/event_rule_baseline/release_pilot_event_001_seed_23/writeup.md`.
2. **Case 2:** `pilot_market_momentum_baseline_release_001`. **Task/system:** `synthetic_market_direction_v0` / `momentum_baseline` (baseline, run label `release_pilot_market_001_seed_11`). **Audit label:** `minimally_defensible`, total score 6. **Low-scoring dimensions:** `temporal_protocol_correctness`, `quantitative_evidence_use`, `baseline_comparison_or_counterfactual_context`, `reproducibility_trace_completeness`. **Primary findings:** No quantitative support in the writeup; No baseline or counterfactual comparison; Temporal protocol rationale is thinner than the artifact package itself. **Evidence:** The submission does not compare the momentum rule to any simpler or more expressive alternative. **Artifact reference:** `runs/suites/pilot_baselines_v0/synthetic_market_direction_v0/momentum_baseline/release_pilot_market_001_seed_11/writeup.md`.
3. **Case 3:** `pilot_event_rule_env_agent_protocol_001`. **Task/system:** `synthetic_event_response_v0` / `event_rule_env_agent` (agent, run label `release_protocol_agent_event_001_seed_23`). **Audit label:** `minimally_defensible`, total score 7. **Low-scoring dimensions:** `temporal_protocol_correctness`, `quantitative_evidence_use`, `baseline_comparison_or_counterfactual_context`. **Primary findings:** Agent trace provenance is stronger than the baseline trace; Narrative quality is still thin; No quantitative or comparative justification appears in the writeup. **Evidence:** The agent output does not situate the rule against any competing event heuristic. **Artifact reference:** `runs/suites/pilot_protocol_v0/synthetic_event_response_v0/event_rule_env_agent/release_protocol_agent_event_001_seed_23/writeup.md`.
4. **Case 4:** `pilot_market_logistic_baseline_release_001`. **Task/system:** `synthetic_market_direction_v0` / `logistic_regression_baseline` (baseline, run label `release_pilot_market_001_seed_11`). **Audit label:** `strong_but_incomplete`, total score 10. **Low-scoring dimensions:** `baseline_comparison_or_counterfactual_context`. **Primary findings:** Method description is strong; Quantitative support is present; Main remaining gap is missing comparison against simpler alternatives. **Evidence:** Even though a momentum baseline exists in the suite, the writeup does not compare against it or justify model complexity relative to a rule baseline. **Artifact reference:** `runs/suites/pilot_baselines_v0/synthetic_market_direction_v0/logistic_regression_baseline/release_pilot_market_001_seed_11/writeup.md`.

9 Reproducibility and Release Artifacts

The release build is intentionally scriptable. It rebuilds suite reports, reference results, statistical artifacts, paper artifacts, task cards, data manifests, and the canonical release manifest. A separate smoke check builds the pilot release twice in isolated roots and compares deterministic publication-facing outputs by content digest.

```
PYTHONPATH=src python scripts/build_pilot_release.py \
  --repeat 3 \
  --market-seed 11 \
  --event-seed 23 \
  --treasury-seed 29 \
  --curve-seed 31 \
```

```
--curve3mo-seed 33 \
--front-end-seed 31 \
--usd-seed 37 \
--treasury-snapshot-date 2026-06-21 \
--curve-snapshot-date 2026-06-21 \
--curve3mo-snapshot-date 2026-06-21 \
--front-end-snapshot-date 2026-06-21 \
--usd-snapshot-date 2026-06-21 \
--clean-existing-outputs
```

10 Limitations

- The current pilot uses a small repeated-run count, so uncertainty artifacts should be read as transparency and regression checks rather than definitive significance claims.
- Bundled agents are controlled example wrappers, not a broad external-agent leaderboard.
- The manual-audit layer still needs independent second-reviewer completion before submission-strength claims.
- The task set is strong enough for a workshop pilot, but a top benchmark/dataset venue version should scale to a larger frozen suite with hidden temporal holdouts.

11 Conclusion

FinDS-AgentBench frames financial data-science agent evaluation as a joint test of predictive performance, temporal validity, leakage resistance, reproducibility, and claim discipline. The pilot release provides the first reproducible scaffold for this evaluation and establishes the artifacts needed for an arXiv/workshop submission.

A Full Repeated-Run Uncertainty Table

B Full Pilot Protocol Reference Table

References

[1] Jun Shern Chan, Neil Chowdhury, Oliver Jaffe, James Aung, Dane Sherburn, et al. Mle-bench: Evaluating machine learning agents on machine learning engineering. *arXiv preprint arXiv:2410.07095*, 2024.

[2] Hui Chen, Miao Xiong, Yujie Lu, Wei Han, Ailin Deng, et al. Mlr-bench: Evaluating ai agents on open-ended machine learning research. *arXiv preprint arXiv:2505.19955*, 2025.

[3] Ziru Chen, Shijie Chen, Yuting Ning, Qianheng Zhang, Boshi Wang, et al. Scienceagentbench: Toward rigorous assessment of language agents for data-driven scientific discovery. *arXiv preprint arXiv:2410.05080*, 2024.

[4] Xueyu Hu, Ziyu Zhao, Shuang Wei, Ziwei Chai, Qianli Ma, et al. Infiagent-dabench: Evaluating agents on data analysis tasks. *arXiv preprint arXiv:2401.05507*, 2024.

[5] Qian Huang, Jian Vora, Percy Liang, and Jure Leskovec. Mlagentbench: Evaluating language agents on machine learning experimentation. *arXiv preprint arXiv:2310.03302*, 2023.

[6] Pranab Islam, Anand Kannappan, Douwe Kiela, Rebecca Qian, Nino Scherrer, and Bertie Vidgen. Financebench: A new benchmark for financial question answering. *arXiv preprint arXiv:2311.11944*, 2023.

[7] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. Swe-bench: Can language models resolve real-world github issues? *arXiv preprint arXiv:2310.06770*, 2023.

[8] Srivatsa Kundurthy, Clara Na, Colton Moraine, Anoushka Mohta, Case Winter, et al. Bluefin: Benchmarking llm agents on financial spreadsheets. *arXiv preprint arXiv:2605.30907*, 2026.

[9] Yuhang Lai, Chengxi Li, Yiming Wang, Tianyi Zhang, Ruiqi Zhong, et al. Ds-1000: A natural and reliable benchmark for data science code generation. *arXiv preprint arXiv:2211.11501*, 2022.

[10] Xiangyu Li, Yawen Zeng, Xiaofen Xing, Jin Xu, and Xiangmin Xu. Profit mirage: Revisiting information leakage in llm-based financial agents. *arXiv preprint arXiv:2510.07920*, 2025.

[11] Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, et al. Agentbench: Evaluating llms as agents. *arXiv preprint arXiv:2308.03688*, 2023.

[12] Jiaxuan Lu, Kong Wang, Yemin Wang, Qingmei Tang, Hongwei Zeng, et al. Fintoolbench: Evaluating llm agents for real-world financial tool use. *arXiv preprint arXiv:2603.08262*, 2026.

[13] Muyan Weng, Defu Cao, Wei Yang, Yashaswi Sharma, and Yan Liu. Temporalbench: A benchmark for evaluating llm-based agents on contextual and event-informed time series tasks. *arXiv preprint arXiv:2602.13272*, 2026.

[14] Qianqian Xie, Weiguang Han, Zhengyu Chen, Ruoyu Xiang, Xiao Zhang, et al. Finben: A holistic financial benchmark for large language models. *arXiv preprint arXiv:2402.12659*, 2024.

[15] Thomson Yen, Julian Poeltl, Harshith Srinivas Gear, Yilin Meng, Joshua Fan, et al. Workstreambench: Evaluating llm agents on end-to-end spreadsheet tasks in finance. *arXiv preprint arXiv:2605.22664*, 2026.

[16] Jie Zhu, Yimin Tian, Boyang Li, Kehao Wu, Zhongzhi Liang, et al. Finmcp-bench: Benchmarking llm agents for real-world financial tool use under the model context protocol. *arXiv preprint arXiv:2603.24943*, 2026.

Task	System	Type	n	Mean	95% CI
Front-End Spread	Extra Trees	Baseline	3	0.509	[0.475, 0.542]
Front-End Spread	LogReg	Baseline	3	0.543	[0.543, 0.543]
Front-End Spread	Prev-Day	Baseline	3	0.540	[0.540, 0.540]
Front-End Spread	Rand. Forest	Baseline	3	0.511	[0.492, 0.529]
Front-End Spread	Front-End Sweep Env-Agent	Agent	3	0.543	[0.543, 0.543]
Synthetic Event	Event Rule	Baseline	3	0.637	[0.611, 0.664]
Synthetic Event	Event Rule Env-Agent	Agent	3	0.637	[0.611, 0.664]
Synthetic Market	LogReg	Baseline	3	0.514	[0.475, 0.554]
Synthetic Market	Market Sweep Env-Agent	Agent	3	0.511	[0.461, 0.562]
Synthetic Market	Momentum	Baseline	3	0.521	[0.475, 0.567]
USD AFE vs. EME	Extra Trees	Baseline	3	0.529	[0.523, 0.535]
USD AFE vs. EME	LogReg	Baseline	3	0.500	[0.500, 0.500]
USD AFE vs. EME	Prev-Day	Baseline	3	0.511	[0.511, 0.511]
USD AFE vs. EME	Rand. Forest	Baseline	3	0.512	[0.481, 0.543]
USD AFE vs. EME	USD Sweep Env-Agent	Agent	3	0.507	[0.490, 0.524]
USD Broad	Extra Trees	Baseline	3	0.500	[0.493, 0.506]
USD Broad	LogReg	Baseline	3	0.507	[0.507, 0.507]
USD Broad	Prev-Day	Baseline	3	0.510	[0.510, 0.510]
USD Broad	Rand. Forest	Baseline	3	0.500	[0.485, 0.515]
USD Broad	USD Sweep Env-Agent	Agent	3	0.502	[0.488, 0.515]
Treasury 10Y-2Y Curve	Extra Trees	Baseline	3	0.501	[0.492, 0.510]
Treasury 10Y-2Y Curve	LogReg	Baseline	3	0.530	[0.530, 0.530]
Treasury 10Y-2Y Curve	Prev-Day	Baseline	3	0.500	[0.500, 0.500]
Treasury 10Y-2Y Curve	Rand. Forest	Baseline	3	0.498	[0.489, 0.506]
Treasury 10Y-2Y Curve	Treasury 10Y-2Y Sweep Env-Agent	Agent	3	0.501	[0.492, 0.510]
Treasury 10Y-3M Curve	Extra Trees	Baseline	3	0.533	[0.505, 0.561]
Treasury 10Y-3M Curve	LogReg	Baseline	3	0.517	[0.517, 0.517]
Treasury 10Y-3M Curve	Prev-Day	Baseline	3	0.500	[0.500, 0.500]
Treasury 10Y-3M Curve	Rand. Forest	Baseline	3	0.532	[0.513, 0.552]
Treasury 10Y-3M Curve	Treasury 10Y-3M Sweep Env-Agent	Agent	3	0.517	[0.517, 0.517]
Treasury 10Y Direction	Extra Trees	Baseline	3	0.506	[0.487, 0.526]
Treasury 10Y Direction	LogReg	Baseline	3	0.487	[0.487, 0.487]
Treasury 10Y Direction	Prev-Day	Baseline	3	0.494	[0.494, 0.494]
Treasury 10Y Direction	Rand. Forest	Baseline	3	0.503	[0.482, 0.525]
Treasury 10Y Direction	Treasury 10Y Sweep Env-Agent	Agent	3	0.504	[0.494, 0.514]

Table 3: Pilot repeated-run uncertainty for Overall. Intervals are t-based 95% confidence intervals over completed runs.

Task	Agent	Type	n	Overall	Balanced Acc.	ROC AUC
Front-End Spread	Extra Trees	Baseline	3	0.509 ± 0.013	0.509 ± 0.013	0.529 ± 0.003
Front-End Spread	LogReg	Baseline	3	0.543 ± 0.000	0.543 ± 0.000	0.536 ± 0.000
Front-End Spread	Prev-Day	Baseline	3	0.540 ± 0.000	0.540 ± 0.000	0.540 ± 0.000
Front-End Spread	Rand. Forest	Baseline	3	0.511 ± 0.008	0.511 ± 0.008	0.518 ± 0.003
Front-End Spread	Front-End Sweep Env-Agent	Agent	3	0.543 ± 0.000	0.543 ± 0.000	0.536 ± 0.000
Synthetic Event	Event Rule	Baseline	3	0.637 ± 0.011	0.637 ± 0.011	0.691 ± 0.018
Synthetic Event	Event Rule Env-Agent	Agent	3	0.637 ± 0.011	0.637 ± 0.011	0.691 ± 0.018
Synthetic Market	LogReg	Baseline	3	0.514 ± 0.016	0.514 ± 0.016	0.528 ± 0.028
Synthetic Market	Momentum	Baseline	3	0.521 ± 0.018	0.521 ± 0.018	0.518 ± 0.024
Synthetic Market	Market Sweep Env-Agent	Agent	3	0.511 ± 0.020	0.511 ± 0.020	0.518 ± 0.023
USD AFE vs. EME	Extra Trees	Baseline	3	0.529 ± 0.002	0.529 ± 0.002	0.541 ± 0.002
USD AFE vs. EME	LogReg	Baseline	3	0.500 ± 0.000	0.500 ± 0.000	0.542 ± 0.000
USD AFE vs. EME	Prev-Day	Baseline	3	0.511 ± 0.000	0.511 ± 0.000	0.511 ± 0.000
USD AFE vs. EME	Rand. Forest	Baseline	3	0.512 ± 0.013	0.512 ± 0.013	0.532 ± 0.002
USD AFE vs. EME	USD Sweep Env-Agent	Agent	3	0.507 ± 0.007	0.507 ± 0.007	0.517 ± 0.011
USD Broad	Extra Trees	Baseline	3	0.500 ± 0.003	0.500 ± 0.003	0.490 ± 0.007
USD Broad	LogReg	Baseline	3	0.507 ± 0.000	0.507 ± 0.000	0.495 ± 0.000
USD Broad	Prev-Day	Baseline	3	0.510 ± 0.000	0.510 ± 0.000	0.510 ± 0.000
USD Broad	Rand. Forest	Baseline	3	0.500 ± 0.006	0.500 ± 0.006	0.506 ± 0.006
USD Broad	USD Sweep Env-Agent	Agent	3	0.502 ± 0.005	0.502 ± 0.005	0.497 ± 0.015
Treasury 10Y-2Y Curve	Extra Trees	Baseline	3	0.501 ± 0.004	0.501 ± 0.004	0.522 ± 0.000
Treasury 10Y-2Y Curve	LogReg	Baseline	3	0.530 ± 0.000	0.530 ± 0.000	0.558 ± 0.000
Treasury 10Y-2Y Curve	Prev-Day	Baseline	3	0.500 ± 0.000	0.500 ± 0.000	0.530 ± 0.000
Treasury 10Y-2Y Curve	Rand. Forest	Baseline	3	0.498 ± 0.004	0.498 ± 0.004	0.511 ± 0.001
Treasury 10Y-2Y Curve	Treasury 10Y-2Y Sweep Env-Agent	Agent	3	0.501 ± 0.004	0.501 ± 0.004	0.522 ± 0.000
Treasury 10Y-3M Curve	Extra Trees	Baseline	3	0.533 ± 0.011	0.533 ± 0.011	0.550 ± 0.003
Treasury 10Y-3M Curve	LogReg	Baseline	3	0.517 ± 0.000	0.517 ± 0.000	0.526 ± 0.000
Treasury 10Y-3M Curve	Prev-Day	Baseline	3	0.500 ± 0.000	0.500 ± 0.000	0.511 ± 0.000
Treasury 10Y-3M Curve	Rand. Forest	Baseline	3	0.532 ± 0.008	0.532 ± 0.008	0.540 ± 0.006
Treasury 10Y-3M Curve	Treasury 10Y-3M Sweep Env-Agent	Agent	3	0.517 ± 0.000	0.517 ± 0.000	0.526 ± 0.000
Treasury 10Y Direction	Extra Trees	Baseline	3	0.506 ± 0.008	0.506 ± 0.008	0.525 ± 0.003
Treasury 10Y Direction	LogReg	Baseline	3	0.487 ± 0.000	0.487 ± 0.000	0.486 ± 0.000
Treasury 10Y Direction	Prev-Day	Baseline	3	0.494 ± 0.000	0.494 ± 0.000	0.494 ± 0.000
Treasury 10Y Direction	Rand. Forest	Baseline	3	0.503 ± 0.009	0.503 ± 0.009	0.523 ± 0.004
Treasury 10Y Direction	Treasury 10Y Sweep Env-Agent	Agent	3	0.504 ± 0.004	0.504 ± 0.004	0.527 ± 0.000

Table 4: Combined Pilot Protocol repeated-run summary. Means and sample standard deviations are computed over the configured seeds.